

Ratify + Google ADK

Cross-domain multi-agent authorization at scale

This independent executable reference answers how delegated authority can cross trust domains, survive multi-agent handoffs, and remain enforceable as protected calls grow from 10 to 1,000,000. It is not a Google partnership, endorsement, or production capacity claim.

The decision boundary

ADK selects and routes a tool call. Workload identity authenticates the caller. The receiving service reconstructs the exact consequential operation and verifies Ratify evidence immediately before the protected action. The model never handles keys or proof bytes.

Trust Domain A → ADK coordinator → ADK broker → Trust Domain B worker → independent receiver → protected action

Two roots, one receiver decision

The receiver pins an authority root for the delegated mandate and a separate workload-admission root for the receiving domain. Both chains must recognize the same worker key. A widened region or node ceiling is denied before the protected handler runs.

Scenario	Result	Action
Valid worker + admission	allow	1 invocation
Broker widens ceiling	constraint_denied	0 invocations
Foreign admission root	admission_binding_failed	0 invocations
Revoked or replayed proof	deny	0 invocations

What changes at scale

Every measured call receives a unique challenge, signs the operation, verifies both chains, evaluates constraints and revocation, consumes the challenge, and records the protected action.

Tier	Recorded result
10 / 100 / 1,000	Measured exact-call workload
1,000,000	900,000 allows; 100,000 constraint denials; 900,000 protected actions
Throughput	369.693 receiver decisions/sec with 8 workers
Per-core context	Approximately 154 decisions/sec/logical core
Encoded proof	56,244 bytes, dual-root hybrid presentation

The scale sample deliberately uses constraint denials only. Replay, revocation, malformed-admission, and untrusted-root cases are covered by the deterministic gate, not mixed into the throughput sample.

Production shape

The local reference uses one process. A production deployment distributes trust bundles and signed revocation snapshots, runs stateless verifier workers regionally, and uses a shared atomic challenge store so a single-use challenge cannot be accepted twice. Ratify Verify is not required synchronously on every call.

Measured versus not deployed

Measured: public ADK callback path, Streamable HTTP MCP, dual-root federation, multi-agent routing, receiver-side enforcement, revocation, replay resistance, and the exact-call local benchmark. Not deployed here: Vertex AI Agent Engine, Google Agent Identity, A2A transport, TLS workload authentication, multi-host storage, external revocation distribution, or real Google Cloud provisioning.

Run, inspect, reproduce

The reference runs without a Gemini API key or Google Cloud project.

Deterministic gate

```
./scripts/google-adk-reference-check.sh
```

Latest recorded run: 49 tests passed, zero skipped, failed, or xfailed.

Scale harness

```
python -m authority_reference.scale_benchmark --calls 10,100,1000,1000000 --workers 8 --output  
evidence/federation-scale-local-rerun.json
```

Canonical links

Source: github.com/identities-ai/ratify-protocol/tree/main/references/google-adk

Architecture and scale note: [references/google-adk/FEDERATION_SCALE.md](#)

Evidence: [references/google-adk/evidence/reference-evidence.md](#) and [federation-scale-local.json](#)

Visual walkthrough: labs.ratifyprotocol.com/google-adk

Positioning for platform teams

ADK, IAM, Agent Identity, A2A, and Agent Engine can continue to do what they already do. Ratify adds a portable, receiver-enforced proof that a recognized principal authorized this exact action under these constraints.